LESSON 04 · CHEAT SHEET

Lists & Dictionaries

LIST: A COLUMN

```
heroes = ["Bram", "Wren", "Piran"]
heroes[0]           'Bram'    # first (from 0)
heroes[-1]          'Piran'   # last
heroes[1:3]         ['Wren', 'Piran']
len(heroes)         3
"Wren" in heroes    True
heroes.append("Hild") # add at the end
heroes.remove("Bram") # remove by value
heroes.pop()        # remove & return last
heroes.index("Wren") # position of
[]                  # empty list
```

NUMBERS IN A LIST

```
r = [320, 1240, 150]
sum(r)  max(r)  min(r)  len(r)
sum(r) / len(r)              # average
sorted(r)                    # sorted COPY
sorted(r, reverse=True)[:3]  # top 3
r.sort()                     # in place, returns None
```

BUILD A LIST WITH A LOOP

```
big = []                     # before
for x in rewards:
    if x > 500:
        big.append(x)        # inside
print(big)                   # after
```

DICTIONARY: A ROW WITH HEADERS

```
hero = {"name": "Wren", "level": 19}
hero["name"]            'Wren'
hero["level"] = 20      # change
hero["pet"] = "hawk"    # add a field
hero["gold"]            # KeyError!
hero.get("gold", 0)     0  # safe lookup
"name" in hero          True
len(hero)               3
del hero["pet"]         # remove a field
{}                      # empty dict
```

Keys: exact and case-sensitive. Values: anything.

LOOP OVER A DICTIONARY

```
for key in hero:             # keys
    print(key, hero[key])
for k, v in hero.items():    # both
    print(k, v)
list(hero.keys())  list(hero.values())
sorted(hero)                 # keys, sorted
```

TABLE = LIST OF DICTS

```
party = [
    {"name": "Bram", "level": 11},
    {"name": "Wren", "level": 19},
]
for hero in party:
    print(hero["name"])
f"{hero['name']}"          # other quotes inside

top = party[0]
for hero in party:
    if hero["level"] > top["level"]:
        top = hero
```

COUNTIF / GROUPING

```
count = {}
for c in classes:
    if c in count:
        count[c] += 1
    else:
        count[c] = 1

groups = {}                  # dict of lists
for q in quests:
    key = q["monster"]
    if key not in groups:
        groups[key] = []
    groups[key].append(q["paid"])
```

EXCEL → PYTHON

a column	a list
a row + headers	a dict
a sheet	list of dicts
`XLOOKUP`	`d[key]` / `d.get(key, x)`
`COUNTIF`	counting loop
`SUM/MAX/AVERAGE`	`sum/max`, `sum/len`
Filter	loop + `append`

GOTCHAS

- `.sort()`, `.append()` return `None`
- `d["ironhills"]` ≠ `d["Ironhills"]` : KeyError
- `f"{p["x"]}"` → use `'x'` inside
- `heroes[4]` on 4 items: IndexError (last is 3)
- Python prints lists with single quotes. Normal.